

JUNRAN YANG

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EDUCATION

University of California San Diego | La Jolla, CA

BS in Data Science, BS in Applied Mathematics

Provost Honors (multiple quarters) • GPA: 3.80 / 4.00

USAMO qualified

TECHNICAL SKILLS

Programming: Python, R, SQL, Java

Libraries: Pandas, NumPy, Scikit-learn, PyTorch (basic)

Data Analysis and Engineering: Statistical Inference, Data Cleaning, Feature Engineering, Data Pipelines

Databases: PostgreSQL, SQL Query Optimization

EXPERIENCE

World Model Research Lab – UC Davis | Research Intern 2025 Summer

- Built a data pipeline using Gymnasium and collect state–action–next-state trajectories for model training
- Implemented a PyTorch world model to predict dynamics and developed encoder networks for latent state representations
- Integrated a contrastive learning module by constructing state pairs to improve representation quality
- Ran experiments and debugged training instability by tuning loss weights, validating data flow, and analyzing prediction performance

Baidu, Inc. – Beijing, China | Project Management Intern 2024 Summer

- Validated and cleaned structured datasets to support large-scale technical projects
- Coordinated cross-functional teams and translated technical requirements into documentation
- Assisted in organizing data workflows and maintaining project data consistency

PROJECTS

Recipe Nutrition & Rating Analysis (End-to-End Data Science Project)

- Analyzed 83,000+ Food.com recipes to study relationships between nutrition features and user ratings
- Built regression models to predict calorie and used cross-validation techniques to evaluate performance
- Conducted permutation hypothesis testing to compare rating between meat and non-meat recipes
- Performed bias analysis to assess differences in user ratings across categories
- Engineered features and cleaned large-scale data using Pandas and NumPy to support statistical analysis

Tech: Python, Pandas, NumPy, Scikit-learn

COVID-19 & Household Wealth Inequality Analysis

- Built an interactive data visualization to analyze household net worth trends across demographic groups
- Processed multi-source economic datasets, including adjustments for inflation to compare real vs. nominal
- Conducted time-series and comparative analysis to identify unequal wealth growth patterns
- Designed dynamic visualizations using D3.js to communicate disparities in wealth distribution effectively
- Revealed that while nominal wealth increased post-pandemic, real gains were significantly reduced and unevenly distributed

Tech: JavaScript, D3.js, HTML/CSS, Jupyter

Full project details available on personal website frank-junran-yang.github.io/Personal-Page